

Murtuza Salman Mohammed

Calgary, AB, Canada | +1 437 663 1488 | murtuzasalman88@gmail.com | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Senior Software Engineer with **4+ years** of experience in **full-stack development** and **AI engineering**. Proven track record leading end-to-end project delivery and driving process improvements through scalable, reliable solutions and performance optimization. Skilled in **Java**, **Spring Boot**, and **React**, with a focus on building enterprise-grade applications for maintainability and high availability. Experienced in leveraging **AI-assisted development tools** (Claude, Cursor, GitHub Copilot) to accelerate delivery, improve code quality, and boost engineering productivity.

PERSONAL OBJECTIVE

Pursuing a responsible and challenging position in a growth-oriented organization where my qualification, knowledge, extensive technical experience, analytic and professional/personal skills can be better harnessed, shared, and enriched.

PERSONAL SKILLS

Innovative, Pro-Active Systematic Problem-Solving, Critical & Analytical Thinking, Ambitious & Goal-oriented, Handling Stressful Situations, Collaborative Teamwork

CORE COMPETENCIES

Languages: Java, Python, TypeScript, JavaScript, C#

Backend & Frameworks: Spring Boot, Hibernate, JPA, RESTful APIs, Apache Camel, Node.js, Express.js

Databases & Caching: PostgreSQL, Amazon RDS, Azure Cosmos DB, Redis

Cloud & Infrastructure: AWS (EC2, RDS, SageMaker), Azure, Docker

CI/CD & DevOps: GitHub Actions, Azure Pipelines, Jenkins, Azure DevOps, Git

Practices & Testing: Agile, TDD, JUnit, Mockito, Spring Boot Test

Monitoring & Observability: Prometheus, Grafana, Datadog, Spring Boot Actuator

AI-Assisted Development: Claude, Cursor, GitHub Copilot

PM & Documentation Tools: Jira, Trello, Swagger/OpenAPI

AI/ML: TensorFlow, Keras, Scikit-learn

PROFESSIONAL EXPERIENCE

Senior Full Stack & AI Engineer April 2025 – Present

Whiz Solutions Inc. – Calgary, AB, Canada

Developed scalable **backend**, **cloud**, and **AI/ML** solutions using **Java**, **Spring Boot**, **AWS**, **Azure**, **Docker**, and **Python**, with a focus on performance, security, and automation.

- Optimized backend performance through **SQL** query tuning, **Redis** caching, lazy loading, and efficient **JPA/Hibernate** data access, improving API response times by up to **25%**.
- Designed and deployed **cloud-native** applications and **microservices** using **Java**, **Spring Boot**, **Docker**, and **AWS** (EC2, RDS, SageMaker), supporting scalable, highly available production environments.
- Developed **RESTful APIs** and backend services using **Java**, **Spring Boot**, and **JPA/Hibernate**, focusing on performance, security, and scalability.
- Leveraged AI coding assistants (**Claude**, **Cursor**, GitHub Copilot) throughout the development lifecycle to accelerate feature delivery, generate boilerplate, and speed up debugging, cutting development time on routine tasks by an estimated **20–30%**.
- Used Claude and Cursor for code review, refactoring, and writing unit tests, improving code quality and test coverage while reducing manual review effort.

- Applied **prompt engineering** techniques to guide AI tools in generating context-aware, production-ready code aligned with team coding standards and architecture patterns.
- Implemented enterprise integration and asynchronous communication patterns using **Apache Camel**.
- Integrated and optimized relational and **NoSQL** databases, including **PostgreSQL** and **Azure Cosmos DB**, applying indexing, partitioning, and caching techniques.
- Automated **CI/CD** workflows using **Azure DevOps**, **Jenkins**, **Git**, and **Docker**, contributing to a **30% reduction** in deployment-related defects.
- Implemented application observability using **Prometheus** and **Grafana**, improving visibility into API performance and infrastructure health.
- Built and deployed production **ML pipelines** on **Amazon SageMaker** with traffic shifting and automated rollback for reliable model deployments using **Guardrails**.
- Engineered a synthetic **computer-vision** data generation pipeline using **OpenCV** and **Albumentations**, producing **10,000+** annotated training images.
- Developed **reinforcement learning** solutions for autonomous systems using **AWS DeepRacer**, placing **1st among 20 teams** in an AI competition.
- Collaborated with engineers, data scientists, QA, and cloud architects to deliver end-to-end solutions across backend, cloud, and **AI/ML** deployment.

Full-Stack Developer *December 2021 – July 2023*

Pilog India Pvt Ltd, Hyderabad, India.

Developed and modernized **Java** applications and **APIs** for enterprise clients, improving performance, scalability, and security while mentoring developers and supporting **Agile** delivery.

- Designed and developed **Java**-based applications for finance and petrochemical clients.
- Migrated legacy codebase from **JDK 8** to **JDK 17**, improving performance by **40%**.
- Developed **RESTful APIs** using **Spring Boot**, integrating third-party systems and securing calls with **JWT**.
- Implemented load-balancing strategies for distributed applications, improving availability, and traffic distribution.
- Designed scalable database models and data access layers using **Hibernate** and **JPA**, applying transaction management and concurrency controls.
- Implemented pessimistic locking for critical financial transactions, maintaining data consistency and integrity.
- Managed database schema migrations using **Flyway**, automating version-controlled changes across environments.
- Mentored junior developers, conducted code reviews, and contributed to **Agile** adoption.
- Adopted early AI-assisted coding tools (ChatGPT and GitHub Copilot) to speed up API development and reduce boilerplate, improving my development throughput.
- Documented technical specifications, REST endpoints, and architecture diagrams to support maintainability and team handoffs.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science *(Graduated)*

Southern Alberta Institute of Technology, Calgary, AB, Canada.

Artificial Intelligence with Machine Learning *(Graduated)*

Humber College, Toronto, ON, Canada.

Object-Oriented Software Development *(Graduated)*

Muffakham Jah College of Engineering and Technology, Hyderabad, India.